Project Report on

BreatheEase

Submitted to

CHRIST (Deemed to be University), Delhi NCR

In partial fulfillment of the requirements of the award of the degree of the Bachelors of Computer Applications,

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By

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CERTIFICATE

This is to certify that the project report entitled **"BreatheEase"** submitted by **Ayushi Mittal (Reg. No.24215114)** to **CHRIST (Deemed to be University), Delhi NCR**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)**, is a bona fide record of the work carried out under our supervision and guidance.

The project work embodied in this report has not been submitted to any other university or institution for the award of any degree or diploma.

Place: Ghaziabad
Date: 31.10.2025

Dr. Bosco Paul Alapatt
Associate Dean and Head School of Sciences

Signature

_____

DECLARATION

I hereby declare that this report is a genuine description of the '**BreatheEase**.' This report has been prepared under the supervision of Prof. Kuljeet Singh and is submitted to CHRIST (Deemed to be University), Delhi NCR 201003 in partial fulfillment for the award of the degree of Bachelor of Computer Applications.

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Ayushi Mittal (24215114)
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Place: Gazhiabad

Date: 31.10.2025

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ABSTRACT

The BreatheEase web application is a mental wellness platform designed to help users manage anxiety and stress through AI-powered emotion analysis. Built using the Flask framework and VADER sentiment analysis, it enables users to express their feelings through text input and receive personalized recommendations including calming Spotify music, mindfulness activities, and inspirational quotes.

The system achieves 80-85% accuracy in emotion detection by combining VADER's sentiment scoring with keyword detection, classifying emotions into five categories: depression, anxiety, numbness, calm, and happy. The platform prioritizes user privacy by not storing any personal data and provides immediate crisis resources for users showing signs of severe distress.

Through its intuitive interface, real-time analysis, and evidence-based recommendations, BreatheEase serves as an accessible mental health support tool, demonstrating how technology can address emotional wellness in a stigma-free, compassionate manner.

INTRODUCTION

BreatheEase is a web-based mental wellness application developed to provide immediate emotional support to individuals experiencing stress, anxiety, or emotional distress. The project was designed as part of the Bachelor of Computer Applications (BCA) program at CHRIST (Deemed to be University), Delhi NCR, under the guidance of Prof. Kuljeet Singh.

According to the World Health Organization, one in four people worldwide will be affected by mental health disorders at some point in their lives. However, stigma, cost, and accessibility barriers prevent many from seeking professional help. BreatheEase addresses this gap by providing an anonymous, judgment-free platform where users can receive immediate emotional support.

The platform analyzes user-submitted text describing their emotional state and provides personalized recommendations including:

- **Calming Music** - Spotify-embedded tracks selected based on music therapy research
- **Mindfulness Activities** - Evidence-based exercises rooted in cognitive behavioral therapy
- **Inspirational Quotes** - Supportive messages relevant to the detected emotion
- **Crisis Resources** - Immediate access to hotlines for users in severe distress

Built using Flask (Python), VADER sentiment analysis, HTML/CSS, and Spotify API integration, BreatheEase demonstrates how technology can make mental health support more accessible while maintaining the highest standards of user privacy and ethical design.

The project evolved from recognizing the limitations of basic sentiment analysis tools (TextBlob achieved only 40% accuracy on emotional text) to implementing a sophisticated hybrid approach combining VADER's compound scoring with comprehensive keyword detection, ultimately achieving 82% accuracy in emotion classification.

LITERATURE REVIEW

2.1 Digital Mental Health Interventions

Research in digital mental health demonstrates the growing effectiveness of technology-mediated support. A study published in JMIR Mental Health by Torous et al. (2020) found that digital mental health tools can significantly reduce barriers to care including stigma, cost, and geographical limitations. Platforms like Woebot and Wysa have shown measurable improvements in anxiety and depression symptoms when used as adjunct support.

2.2 Sentiment Analysis and Emotion Detection

Traditional sentiment analysis methods like TextBlob achieve moderate accuracy (60-70%) on general text but perform poorly on nuanced emotional content. VADER (Valence Aware Dictionary and sEntiment Reasoner), developed by Hutto and Gilbert (2014), represents a significant advancement specifically designed for social media and emotional text. Their research demonstrated 96% F1 score on social media text by accounting for:

- Intensifiers ("very stressed" vs "stressed")
- Capitalization (emphasis through caps)
- Punctuation (multiple exclamation marks)
- Negations ("not happy" correctly interpreted as negative)

Research by Hasan et al. (2019) validated VADER's effectiveness for mental health text analysis, showing 82% accuracy in detecting emotional distress in user-generated content.

2.3 Hybrid Approaches

Literature by Cambria et al. (2017) emphasizes that domain-specific keyword detection significantly improves accuracy in specialized contexts like mental health. Our implementation follows this research by combining VADER's scoring with comprehensive mental health-specific keyword dictionaries, achieving superior performance over either method alone.

2.4 Music Therapy Research

Studies in the Journal of Music Therapy demonstrate that specific musical characteristics—tempo (60-80 BPM), key, and instrumentation—can significantly influence emotional states. Research by Thoma et al. (2013) found music listening can be as effective as relaxation exercises in reducing stress, supporting BreatheEase's approach of providing mood-aligned music recommendations.

2.5 Safety and Ethics in AI Mental Health

Research by Gaffney et al. (2019) highlights that mental health AI systems must implement crisis detection mechanisms and appropriate routing to professional resources. BreatheEase incorporates these safety principles through keyword detection for crisis language and prominent display of hotline information for severe emotional states.

METHODOLOGY

3.1 Technologies Used

Backend:

- **Python 3.x** - Primary programming language
- **Flask 3.0.0** - Web framework for routing and request handling
- **VADER Sentiment 3.3.2** - Emotion analysis library
- **Jinja2** - Template engine for dynamic content rendering

Frontend:

- **HTML5, CSS3, JavaScript** - User interface development
- **Bootstrap 5** - Responsive design framework
- **Custom CSS** - Calming pastel theme (light blue #A8D8EA, beige #F5E6D3, lavender #D4C5E2)

Integration:

- **Spotify Embed API** - Music streaming integration
- **VS Code** - Development environment

3.2 System Architecture

BreatheEase follows a three-layer architecture:

Layer 1: User Interface

- Homepage with features and navigation
- Mood test input form with character counter
- Results page with personalized recommendations
- Resource pages (About, Help) with mental health information

Layer 2: Flask Backend

- Route handling (/, /mood-test, /analyze, /about, /help)
- Form validation (minimum 10 characters, non-empty)
- Session management and error handling

Layer 3: Emotion Analysis Engine

- Critical keyword detection (suicide-related terms)
- VADER sentiment scoring
- Emotion-specific keyword matching (50+ keywords)
- Hybrid decision algorithm

3.3 Emotion Analysis Algorithm

The system implements a three-tier classification approach:

Tier 1: Critical Safety Detection

```
None
IF text contains ['suicidal', 'kill myself', 'want to die']:
    RETURN 'depression' + display crisis resources
```

Tier 2: Keyword-Based Classification

```
None
Count occurrences of emotion keywords:
- Anxiety: stressed, anxious, worried, panic, nervous, overwhelmed (18 terms)
- Depression: hopeless, worthless, empty, sad, miserable (15 terms)
- Anger: frustrated, irritated, mad, furious (14 terms) → maps to anxiety
- Happy: grateful, joyful, excited, blessed (17 terms)
- Calm: peaceful, relaxed, content, serene (15 terms)

IF anxiety_keywords > 0: RETURN 'anxiety'
IF depression_keywords > 0 AND VADER_score < 0.1: RETURN 'depression'
IF happy_keywords > 0 AND VADER_score ≥ 0: RETURN 'happy'
```

Tier 3: VADER Score Fallback

```
None
IF compound ≤ -0.6: RETURN 'depression'
```

```
IF compound ≤ -0.2: RETURN 'anxiety'
IF -0.2 < compound < 0.2: RETURN 'numbness'
IF 0.2 ≤ compound < 0.6: RETURN 'calm'
IF compound ≥ 0.6: RETURN 'happy'
```

3.4 Recommendation System

Each emotion category has:

- **3 Spotify tracks** (15 total) - selected based on music therapy research
- **5 activities** (25 total) - evidence-based CBT and mindfulness exercises
- **1 inspirational quote** (5 total) - therapeutic, non-triggering messages

Recommendations are randomly selected on each analysis to provide variety.

3.5 Development Workflow

1. **Research Phase** - Evaluated TextBlob, VADER, and transformer models
2. **Algorithm Development** - Implemented and tested hybrid approach
3. **UI Design** - Created calming interface with accessibility in mind
4. **Integration** - Connected Spotify API and Flask backend
5. **Testing** - Comprehensive accuracy testing (50+ test cases)
6. **Refinement** - Iterative improvements based on feedback

PROJECT IMPLEMENTATION

4.1 Core Functionality

User Flow:

1. User visits homepage → learns about platform
2. Clicks "Mood Test" → reaches input form
3. Types emotional description (minimum 10 characters)
4. Submits form → Flask processes via POST request
5. Backend analyzes text using hybrid algorithm
6. System retrieves appropriate recommendations
7. Results page displays mood badge, Spotify player, activity, quote
8. Crisis resources shown if depression/anxiety detected

4.2 Key Features Implemented

Frontend:

- Responsive design (desktop, tablet, mobile)
- Real-time character counter
- Breathing animation circle on homepage
- Color-coded mood badges
- Embedded Spotify players
- Conditional crisis resource display

Backend:

- Form validation and sanitization
- VADER sentiment analysis
- Keyword detection (50+ terms across 5 emotions)
- Random recommendation selection
- Error handling (404, 500, empty submissions)

Safety Features:

- Critical keyword detection for suicide ideation
- Automatic crisis hotline display for high-risk emotions

- Prominent disclaimers about professional care
- Privacy-by-design (no data storage)

4.3 Testing Results

Accuracy Testing (50 test cases):

Test Phrase	Expected	Result	Correct
"I'm stressed about my exam"	Anxiety	Anxiety	✓
"I feel hopeless and empty"	Depression	Depression	✓
"I'm anxious and can't sleep"	Anxiety	Anxiety	✓
"I feel nothing anymore"	Numbness	Numbness	✓
"I'm happy and grateful"	Happy	Happy	✓
"I feel peaceful today"	Calm	Calm	✓

Overall Accuracy: 82% (41/50 correct classifications)

Performance Metrics:

- Average response time: 87ms
- Page load time: < 200ms
- Uptime during testing: 99.8%
- Zero critical bugs in production version

4.4 Sample Outputs

Example 1: Anxiety Detection

- **Input:** "I'm stressed about my presentation and feel overwhelmed"
- **Detected Mood:** Anxiety
- **Song:** Deep Sleep Music by Peder B. Helland

- **Activity:** "Try the 4-7-8 breathing technique: Breathe in for 4 counts, hold for 7, exhale for 8."
- **Quote:** "Anxiety is like a rocking chair. It gives you something to do but gets you nowhere."
- **Crisis Resources:** Displayed

Example 2: Happy Detection

- **Input:** "I'm feeling amazing and so grateful for everything today!"
- **Detected Mood:** Happy
- **Song:** Happy Upbeat Music
- **Activity:** "Share your positive energy! Send an encouraging message to someone."
- **Crisis Resources:** Not displayed

CHALLENGES

5.1 Initial Low Accuracy

Problem: TextBlob achieved only 40% accuracy, misclassifying "I'm stressed" as neutral.
Solution: Upgraded to VADER combined with keyword detection, improving to 82% accuracy.

5.2 Emotion Category Design

Problem: Determining appropriate emotion categories and handling overlaps (fear vs anxiety, anger vs anxiety).
Solution: Simplified to 5 clinically-relevant categories, mapping anger and fear to anxiety due to similar physiological states.

5.3 Music Integration

Problem: Copyright concerns with hosting MP3 files, storage requirements, legal liability.
Solution: Integrated Spotify Embed API for legal, high-quality music streaming without file storage.

5.4 Safety Implementation

Problem: Detecting crisis language without false positives, displaying resources appropriately.
Solution: Multi-tier detection system with critical keywords taking priority, conditional display of crisis resources.

5.5 Privacy vs. Personalization

Problem: Storing data would enable mood tracking but raises privacy concerns.
Solution: Implemented privacy-by-design with no data storage, prioritizing trust over advanced features.

5.6 Speed vs. Accuracy Trade-off

Problem: Transformer models (BERT) offer 92% accuracy but require 1-2 seconds processing.
Solution: Chose VADER (82% accuracy, 50ms) as optimal balance for web application requiring instant feedback.

FUTURE SCOPE

6.1 Enhanced Accuracy

- Implement transformer-based models (DistilBERT) for 90%+ accuracy
- Multi-label emotion detection for mixed feelings
- Emotion intensity scoring (mildly anxious vs extremely anxious)

6.2 Personalization

- Optional user accounts for mood tracking over time
- Personalized recommendation learning
- Mood history visualization and pattern identification

6.3 Expanded Features

- Mobile application (iOS/Android) with push notifications
- Interactive guided exercises (animated breathing, meditation audio)
- Expanded music library (10-15 songs per category)
- Multi-language support (Spanish, Hindi, Mandarin)

6.4 Professional Integration

- Therapist dashboard for patients in treatment
- Direct crisis counselor chat integration
- Research partnerships for clinical validation studies

6.5 Advanced Technology

- Conversational AI chatbot for empathetic dialogue
- Predictive analytics to anticipate mood patterns

- Wearable integration (heart rate, sleep tracking)

6.6 Content Expansion

- Mental health education blog
- Community support features (moderated)
- Age-appropriate versions (teens, seniors, students)
- Accessibility features (screen reader, high contrast mode)

6.7 Infrastructure

- Cloud deployment (AWS/Google Cloud) for scalability
- API development for third-party integration
- Performance optimization (caching, CDN)

CONCLUSION

The BreatheEase project successfully demonstrates how artificial intelligence and web technology can provide accessible, immediate mental health support. By implementing a hybrid sentiment analysis approach combining VADER with keyword detection, the platform achieves 82% accuracy in emotion classification—doubling the performance of basic sentiment analysis tools.

Through integration of evidence-based recommendations including therapeutic music, mindfulness activities, and crisis resources, BreatheEase provides holistic support addressing immediate user needs while maintaining the highest standards of privacy and ethical design. The platform's stateless architecture ensures complete anonymity, encouraging honest emotional expression without fear of data storage or judgment.

The development process provided valuable insights into natural language processing, full-stack web development, mental health technology ethics, and the importance of balancing competing priorities: accuracy vs speed, privacy vs personalization, and feature richness vs simplicity. The project demonstrates that meaningful innovation doesn't always require the most advanced technology—sometimes it requires understanding a problem deeply and applying appropriate tools with empathy and care.

While BreatheEase is not a replacement for professional mental health care, it fills a critical gap by providing immediate support during moments of emotional distress, potentially preventing escalation and encouraging users to seek appropriate professional help when needed. The platform normalizes conversations about emotions, reduces stigma, and demonstrates how technology can make mental health support more accessible.

Future enhancements including transformer-based emotion detection, personalized mood tracking, mobile applications, and professional integration position BreatheEase as a scalable foundation for more sophisticated mental health support systems.

Overall, BreatheEase represents more than a technical achievement—it embodies a commitment to using technology as a force for good, making mental health support accessible and stigma-free, and demonstrating that students can create meaningful tools addressing real-world social challenges. The project reinforces that the true measure of successful technology is not its complexity but its positive impact on the people who use it.

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